Option 2A: Glaciated Landscapes and Change

Overview

Ice sheets and glaciers operate within a landscape system as glacial processes of erosion, transport and deposition combine with meteorological and climatological processes and interact with geological and lithological processes to produce distinctive landscapes. The landscapes can be both present day and relict and can occur in both upland and lowland areas. These landscapes are being changed by both physical processes and human activities which pose unique threats due to the low level of resilience found in these areas. Study must include examples of landscapes from areas inside and outside the UK.

Content

Enquiry question 1: How has climate change influenced the formation of glaciated landscapes over time?			
Key idea		**Detailed content**	
2A.1	The causes of longer and shorter climate change from the start of the Pleistocene, through the Holocene and into the Anthropocene.	a.	A chronology of multiple glacial and interglacial periods caused by climate change since the start of the Pleistocene.
		b.	The factors leading to climate change in the Pleistocene, Holocene and Anthropocene: Milankovitch cycles, variations in solar output (1), variations in composition of atmospheric gases, and volcanic eruptions.
2A.2	Present and past Pleistocene distribution of ice cover.	a.	The definition and importance of the cryosphere and its role in global systems and classification of ice masses by scale and location (ice sheets, ice caps, cirque and valley glaciers, and ice fields) and polar and temperate environments.
		b.	The present-day distribution of high latitude ice sheets and evidence for Pleistocene ice sheet extent.
		c.	The present-day distribution of high altitude glaciated upland landscapes and evidence of relict landscapes from the Pleistocene. (2)
2A.3	Periglacial processes produce distinctive landscapes.	a.	Distribution of past and present periglacial landscapes which are underlain by continuous, discontinuous and sporadic areas of permafrost with a seasonally active layer.
		b.	Periglacial processes include nivation, frost heave, freeze-thaw weathering and solifluction as well as high winds and meltwater erosion.
		c.	The formation of often unique periglacial landforms (ice wedges, patterned ground, pingos, loess) contributes towards the occurrence of distinctive periglacial landscapes (Tundra environments of northern Russia or northern Canada).

Enquiry question 2: What processes operate within glacier systems?			
Key idea		**Detailed content**	
2A.4	Mass balance is important in understanding glacial dynamics and the operation of glaciers as systems.	a.	Glacial mass balance system and the relationship between accumulation and ablation in the maintenance of equilibrium. (3) The importance of positive and negative feedback (Greenland Ice Sheet).
		b.	The process of accumulation (direct snowfall, avalanches and wind deposition) and the process of ablation (melting, sublimation, calving, evaporation and avalanches).
		c.	The reasons for the variations in the rates of accumulation and ablation, and the impact these variations have on the mass balance over different timescales.
2A.5	Different processes explain glacial movement and variations in rates.	a.	Polar and temperate glaciers have different rates of movement.
		b.	There are different processes that are important in the movement of glaciers (basal slip, regelation creep, internal deformation).
		c.	A number of factors control the rate of movement (altitude, basal temperature, slope, lithology, size and variations in mass balance) with both positive and negative feedback in the system. (4)
2A.6	The glacier landform system.	a.	Glaciers alter landscapes by a number of processes: details of erosion, entrainment, transport and deposition.
		b.	Glacial landforms develop at macro-, meso- and micro-scales with distinctive morphologies in process environments, such as sub-glacial, marginal, proglacial and periglacial.
		c.	These landforms create a number of distinctive landscapes in upland and lowland areas that can be used to study the extent of ice cover.

Enquiry question 3: How do glacial processes contribute to the formation of glacial landforms and landscapes?			
Key idea		**Detailed content**	
2A.7	Glacial erosion creates distinctive landforms and contributes to glaciated landscapes.	a.	Glacial erosional processes (abrasion, quarrying, plucking, crushing and basal melting, combined with subaerial freeze thaw and mass movement).
		b.	The processes leading to the formation of landforms associated with cirque and valley glaciers (cirques/corries (5), arêtes, pyramidal peaks, glacial troughs, truncated spurs/hanging valleys and ribbon lakes).
		c.	The formation of landforms due to ice sheet scouring (roches moutonnées, knock and lochan, crag and tail) and the influence of differential geology.
2A.8	Glacial deposition creates distinctive landforms and contributes to glaciated landscapes.	a.	The formation of glacial (ice contact) depositional features (medial, lateral, recessional and terminal moraines and drumlins).
		b.	The formation of lowland depositional features (till plains, lodgement and ablation till). (6)
		c.	The assemblage of landforms can be used to reconstruct former ice extent, movement and provenance (erratics, moraines, crag and tail, drumlin orientation). (7)
2A.9	Glacial meltwater plays a significant role in creating distinctive landforms and contributes to glaciated landscapes.	a.	The processes of water movement within the glacial system (supraglacial, englacial and sub-glacial flows).
		b.	Glacial and fluvioglacial deposits have different characteristics (stratification, sorting, imbrication and grading). (8)
		c.	The formation of fluvio-glacial landforms; ice contact features (kames, eskers and kame terraces) and proglacial features (sandurs, pro-glacial lakes, meltwater channels, and kettleholes).

Enquiry question 4: How are glaciated landscapes used and managed today?		
Key idea	**Detailed content**	
2A.10 Glacial and periglacial landscapes have intrinsic cultural, economic and environmental value.	a.	Relict and active glaciated landscapes have environmental and cultural value (polar scientific research, wilderness recreation, and spiritual/religious associations). ***(A: attitudes range from exploitation to preservation)***
	b.	Glaciated landscapes are important economically (farming, mining, hydroelectric power, tourism, forestry) to include a study of contrasting environments around the world.
	c.	Glaciated and periglacial landscapes have a unique biodiversity (tundra) and play an important role in the maintenance of natural systems (water and carbon cycles).
2A.11 There are threats facing fragile active and relict glaciated upland landscapes.	a.	Glaciated landscapes face varying degrees of threat from both natural hazards (avalanches and glacial outburst floods) and human activities (leisure and tourism, reservoir construction, urbanisation) (Alpine valleys).
	b.	Human activity can degrade the landscape and fragile ecology of glaciated landscapes (soil erosion, trampling, landslides, deforestation). ***(A: direct actions by players reduce resilience)***
	c.	Climate change is having a major impact on glacial mass balances, which in turn risks disruption of the hydrological cycle (meltwater, river discharge, sediment yield, water quality) (Himalayan Glaciers). (9) ***(A: indirect actions by players alter natural systems)***
2A.12 Threats to glaciated landscapes can be managed using a spectrum of approaches.	a.	Different stakeholders (conservationists, local and regional government, global organisations, NGOs) are involved in managing the challenges posed by glaciated landscapes, using a spectrum of approaches from protection through to sustainable management and multiple economic use (Yosemite Valley). ***(A: actions range from exploitation to preservation)***
	b.	Legislative frameworks are used to protect and conserve landscapes by conservation and management at a variety of scales.
	c.	Climate change makes successful management of these unique and fragile landscapes increasingly challenging, with a need for coordinated approaches at global, national and local scale. ***(F: this risk is creating an uncertain future and needs mitigation and adaptation)***

Guidance for integrating geographical skills for Topic 2 Option 2A

Study must emphasise the use of quantitative geographical skills, including developing observation skills, measurement and geo-spatial mapping skills, together with data manipulation and statistical skills applied for field measurement. Qualitative approaches may be used if appropriate. The following guidance on integrating skills gives suggestions for opportunities to meet these requirements. These skills are **not** exclusive to the topic areas under which they appear; students will need to be able to apply these skills across any suitable topic area throughout their course of study. The full range of skills is outlined in the geographical skills appendix (*Appendix 1*).

(1) Graphical analysis of reconstructed climate versus landform evidence for past glacial/interglacial periods.

(2) Comparison of past and present distribution of glaciated landscapes using global and regional maps.

(3) Use of numerical data to calculate simple mass balance and equilibrium line position; use of GIS to identify main features of glacier types and assess glacier health.

(4) Use of measures of central tendency to compare rates of glacier movement.

(5) Cirque orientation analysis using large-scale maps (OS maps); calculating Spearman's rank correlations of height of basin, size of basin and orientation and commenting on the significance of the correlation.

(6) Till fabric analysis using rose diagrams.

(7) Use of British Geological Society (BGS) glacial drift maps, Ordnance Survey (OS) maps, GIS and fieldwork results to reconstruct past ice extent and ice flow direction.

(8) Use of student t-test to analyse changes in sediment size and shape in outwash plains; central tendency analysis of both glacial and fluvioglacial deposits (comparison of size, shape and degree of sorting of clasts).

(9) Numerical analysis of mean rates of glacial recession in different global regions.

(10) Drumlin morphometry and orientation survey to measure correlation of height, length and elongation ratio. Statistical comparison of two data sets from contrasting locations.

NB Geographical skills should use data and images from areas inside and outside the UK.

Topic 2: Landscape Systems, Processes and Change

Option 2B: Coastal Landscapes and Change

Overview

Coastal landscapes develop due to the interaction of winds, waves and currents, as well as through the contribution of both terrestrial and offshore sources of sediment. These flows of energy and variations in sediment budgets interact with the prevailing geological and lithological characteristics of the coast to operate as coastal systems and produce distinctive coastal landscapes, including those in rocky, sandy and estuarine coastlines. These landscapes are increasingly threatened from physical processes and human activities, and there is a need for holistic and sustainable management of these areas in all the world's coasts. Study must include examples of landscapes from inside and outside the UK.

Content

Enquiry question 1: Why are coastal landscapes different and what processes cause these differences?			
Key idea		**Detailed content**	
2B.1	The coast, and wider littoral zone, has distinctive features and landscapes.	a.	The littoral zone consists of backshore, nearshore and offshore zones, includes a wide variety of coastal types and is a dynamic zone of rapid change.
		b.	Coasts can be classified by using longer term criteria such as geology and changes of sea level or shorter term processes such as inputs from rivers, waves and tides.
		c.	Rocky coasts (high and low relief) result from geology which is resistant to erosive forces of sea, rain and wind, often in a high-energy environment, whereas coastal plain landscapes (sandy and estuarine coasts) are found near areas of low relief and result from supply of sediment from different terrestrial and offshore sources, often in a low-energy environment.
2B.2	Geological structure influences the development of coastal landscapes at a variety of scales.	a.	Geological structure is responsible for the formation of concordant and discordant coasts.
		b.	Geological structure influences coastal morphology: Dalmatian and Haff type concordant coasts and headlands and bays on discordant coasts.
		c.	Geological structure (bedding planes, jointing, dip, faulting, folding) is an important influence on coastal morphology and erosion rates, and also on the formation of cliff profiles and the occurrence of micro-features, e.g. caves (Glamorgan Heritage Coast). (2)
2B.3	Rates of coastal recession and stability depend on lithology and other factors.	a.	Bedrock lithology (igneous, sedimentary, metamorphic) and unconsolidated material (boulder clay) geology are important in understanding rates of coastal recession.
		b.	Differential erosion of alternating strata in cliffs (permeable/impermeable, resistant/less resistant) produces complex cliff profiles and influences recession rates. (3)
		c.	Vegetation is important in stabilising sandy coastlines through dune successional development and salt marsh successional development in estuarine areas.

Enquiry question 2: How do characteristic coastal landforms contribute to coastal landscapes?

Key idea			Detailed content
2B.4	Marine erosion creates distinctive coastal landforms and contributes to coastal landscapes.	a.	Different wave types (constructive/destructive) influence beach morphology and beach sediment profiles, which vary at a variety of temporal scales from short term (daily) through to longer periods (4)
		b.	The importance of erosion processes (hydraulic action, corrosion, abrasion, attrition) and how they are influenced by wave type, size and lithology.
		c.	Erosion creates distinctive coastal landforms (wave cut notches, wave cut platforms, cliffs, the cave-arch-stack-stump sequence).
2B.5	Sediment transport and deposition create distinctive landforms and contribute to coastal landscapes.	a.	Sediment transportation is influenced by the angle of wave attack, the process of longshore drift, tides and currents. (5)
		b.	Transportation and deposition processes produce distinctive coastal landforms (beaches, recurved and double spits, offshore bars, barrier beaches and bars, tombolos and cuspate forelands), which can be stabilised by plant succession.
		c.	The Sediment Cell concept (sources, transfers and sinks) is important in understanding the coast as a system of dynamic equilibrium, with both negative and positive feedback (Portland Bill to Selsey Bill).
2B.6	Subaerial processes of mass movement and weathering influence coastal landforms and contribute to coastal landscapes.	a.	Weathering (mechanical, chemical, biological) is important in sediment production and influences rates of recession.
		b.	Mass movement (blockfall, rotational slumping, landslides) is important on some coasts with weak and/or complex geology.
		c.	Mass movement creates distinctive landforms (rotational scars, talus scree slopes, terraced cliff profiles).

Enquiry question 3: How do coastal erosion and sea level change alter the physical characteristics of coastlines and increase risks?		
Key idea		**Detailed content**
2B.7 Sea level change influences coasts on different timescales.	a.	Longer-term sea level changes result from a complex interplay of factors both eustatic (ice formation/melting, thermal changes) and isostatic (post glacial adjustment, subsidence, accretion and tectonics).
	b.	Sea level change has produced emergent coastlines (raised beaches with fossil cliffs) and submergent coastlines (rias, fjords and Dalmatian). (6)
	c.	Contemporary sea level change from global warming or tectonic activity is a risk to some coastlines.
2B.8 Rapid coastal retreat causes threats to people at the coast.	a.	Rapid coastal recession is caused by physical factors (geological and marine) but can be influenced by human actions (dredging or coastal management) (the Nile Delta or Guinea coastline or Californian coastline). ***(A: actions of different players may alter natural systems)***
	b.	Subaerial processes (weather and mass movement) work together to influence rates of coastal recession.
	c.	Rates of recession are not constant and are influenced by different factors both short- and longer term (wind direction/fetch, tides, seasons, weather systems and occurrence of storms). (7)
2B.9 Coastal flooding is a significant and increasing risk for some coastlines.	a.	Local factors increase flood risk on some low-lying and estuarine coasts (height, degree of subsidence, vegetation removal); global sea level rise further increases risk (Bangladesh or the Maldives).
	b.	Storm surge events can lead to severe coastal flooding with dramatic short-term impacts (depressions, tropical cyclones).
	c.	Climate change may increase coastal flood risk (frequency and magnitude of storms, sea level rise) but the pace and magnitude of this threat is uncertain. ***(F: this risk is creating an uncertain future and needs mitigation and adaptation)***

Enquiry question 4: How can coastlines be managed to meet the needs of all players?		
Key idea		**Detailed content**
2B.10 Increasing risks of coastal recession and coastal flooding have serious consequences for affected communities.	a.	Economic losses (housing, businesses, agricultural land, infrastructure) and social losses (relocation, loss of livelihood, amenity value) from coastal recession can be significant, especially in areas of dense coastal developments.
	b.	Coastal flooding and storm surge events can have serious economic and social consequences for coastal communities in both developing and developed countries.
	c.	Climate change may create environmental refugees in coastal areas.
2B.11 There are different approaches to managing the risks associated with coastal recession and flooding.	a.	Hard engineering approaches (groynes, sea walls, rip rap, revetments, offshore breakwaters) are economically costly and directly alter physical processes and systems. (8) ***(A: actions by different players may have unforeseen consequences)***
	b.	Soft engineering approaches (beach nourishment, cliff re-grading and drainage, dune stabilisation) attempt to work with physical systems and processes to protect coasts and manage risks caused by changes in sea-level. (9)
	c.	Sustainable management is designed to cope with future threats (increased storm events, rising sea levels) but its implementation can lead to local conflicts in many countries. ***(F: mitigation and adaptation will both be needed for future stability)***
2B.12 Coastlines are now increasingly managed by holistic integrated coastal zone management (ICZM).	a.	Coastal management increasingly uses the concept of littoral cells to manage extended areas of coastline. Throughout the world, countries are developing schemes that are sustainable and use holistic ICZM strategies.
	b.	Shoreline Management Policy decisions (No Active Intervention, Strategic Realignment, Hold The Line, Advance The Line) are based on complex judgements (engineering feasibility, environmental sensitivity, land value, political and social reasons); Cost Benefit Analysis (CBA) and Environmental Impact Assessment (EIA) are used as part of the decision-making process.
	c.	Policy decisions can lead to conflicts between different players (homeowners, local authorities, environmental pressure groups) with perceived winners and losers in countries at different levels of development (developed and developing or emerging countries) (Happisburgh and Chattogram). ***(A: attitudes of differing players may vary)***

Guidance for integrating geographical skills for Topic 2 Option 2B

Study must emphasise the use of quantitative geographical skills, including developing observation skills, measurement and geo-spatial mapping skills, together with data manipulation and statistical skills applied for field measurement. Qualitative approaches may be used if appropriate. The following guidance on integrating skills gives suggestions of opportunities to meet these requirements. These skills are **not** exclusive to the topic areas under which they appear; students will need to be able to apply these skills across any suitable topic area throughout their course of study. The full range of skills is outlined in the geographical skills appendix (*Appendix 1*).

(1) GIS mapping of the variety of coastal landscapes, both for and beyond the UK.

(2) Satellite interpretation of a variety of coastlines to attempt to classify them.

(3) Field sketches of contrasting coastal landscapes.

(4) Using measures of central tendency to classify waves into destructive and constructive wave types.

(5) Using student t-test to investigate changes in pebble size and shape along a drift aligned beach and also across the littoral zone to above the storm beach.

(6) Map and aerial interpretation of distinctive landforms indicating past of sea level change.

(7) Use of GIS, aerial photos and maps to calculate recession rates for a variety of temporal rates (annual changes and longer-term changes).

(8) Interrogation of GIS of management cells to ascertain land use values and develop cost/benefit analysis to inform the choice of coastal management strategy.

(9) Photo interpretation of a range of approaches to management to assess environmental impact.

(10) Sand dune or salt marsh surveys to assess the impact of succession using an index of diversity, X^2 (Chi-square to compare features of the various zones).